ERP Enterprise Pro

Project presentation

Web-based management system: master data, sales, inventory, finance, reports and administration — with the ADVPL equivalent for Protheus.

October 2026

Demo environment · fictitious data

What we'll cover

01 Context and goal

02 The project in numbers

03 How the system is built

04 Access profiles

05 The life of an order

06 The system, step by step

07 Results with demo data

08 Security and quality

09 Limitations and next steps

Context and goal

Build a working web ERP, inspired by Protheus (TOTVS) processes, that demonstrates a manufacturer's sales cycle end to end — and deliver the equivalent ADVPL source code for deployment.

What

A complete web system with 11 pages, business rules on the server and its own database.

Who it's for

Sales, inventory, finance, management and administration — each sees only what they need.

Outcome

A navigable, documented and tested demo that runs on any Windows computer.

The project in numbers

11 pages in the system	**6** access profiles	**49** automated tests passing
10 ADVPL source files for Protheus	**0** external libraries installed	**8.8k** lines of code

Counts taken from the code and the test suite on Oct 21, 2026.

How the system is built

LAYER 1

Browser

The system's screens. Works on desktop and mobile, with nothing to install.

→ requests via API

LAYER 2

Server

Business rules, permissions, validation and auditing. Nothing is decided on the screen alone.

→ transactions

LAYER 3

Database

Customers, products, orders, inventory, receivables and payables, and the audit trail.

Protheus equivalent

The same processes were written in ADVPL (10 source files: customers, products, orders, inventory, finance, reports and REST API), using the ERP's standard routines. **These source files have not been compiled yet** — they must be validated in a staging environment.

Each profile sees and does only what it needs

	Admin	Manager	Sales rep	Inventory	Finance	Read-only
Customers	●	●	●	○	○	○
Products	●	●	○	●	○	○
Orders	● approves & invoices	● approves & invoices	● own orders only	—	○	○
Inventory	● moves stock	○	○	● moves stock	○	○
Finance	● settles	○	—	—	● settles	○
Reports	○	○	○	○	○	○
Audit log	○	○	—	—	—	—
Users & admin panel	●	—	—	—	—	—

● create & edit ○ view only — no access

Rules are enforced on the server, not just hidden on screen.

The life of an order

1
Draft
The sales rep picks the customer and items. Freely editable.

→

2
Submit
The system checks stock, discount and credit limit.

→

3
! Pending approval
Only when something exceeds a limit. The manager approves or rejects.

→

4
◆ Approved
Stock is reserved for this order.

→

5
✓ Invoiced
Deducts stock and creates the receivables.

Automatic rules

Discounts up to 10% need no approval; up to 25% with manager approval. Orders above the customer's credit go to approval.

Control

Whoever creates an order cannot approve it. Every step is logged with user, date and reason.

Alternative outcomes

✕ Rejected (back for changes) or – Cancelled (releases the reservation; cannot be undone).

System access

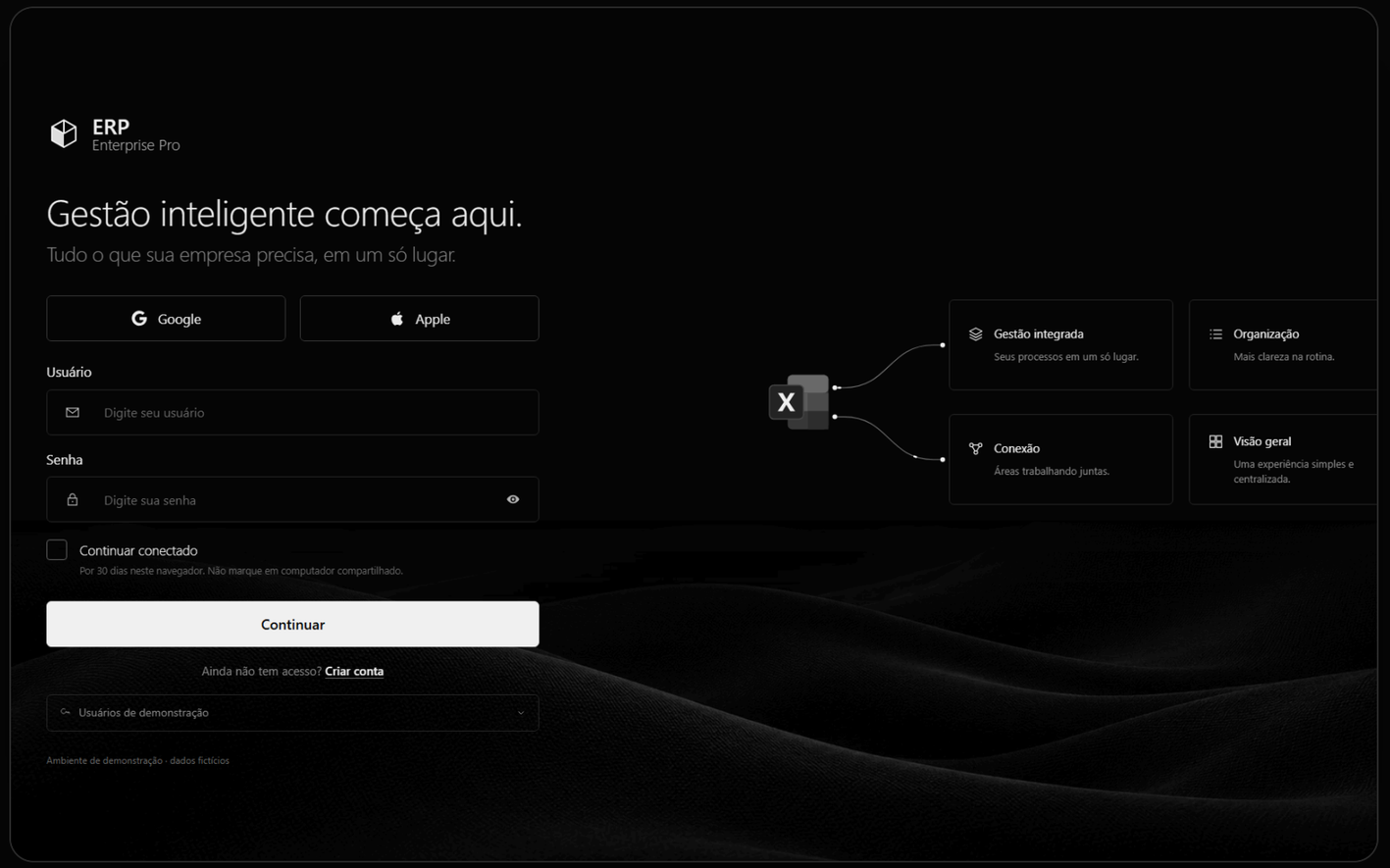
Sign in with username and password; anyone without an account can sign up.

1. **Sign in**
   Case-insensitive username; eye icon to check the password.

2. **Stay signed in**
   Keeps the session for 30 days on the user's own computer.

3. **Create account**
   New accounts are read-only; the administrator grants the rest.

4. **Protection**
   After 5 wrong passwords, sign-in is locked for 60 seconds.



Overview

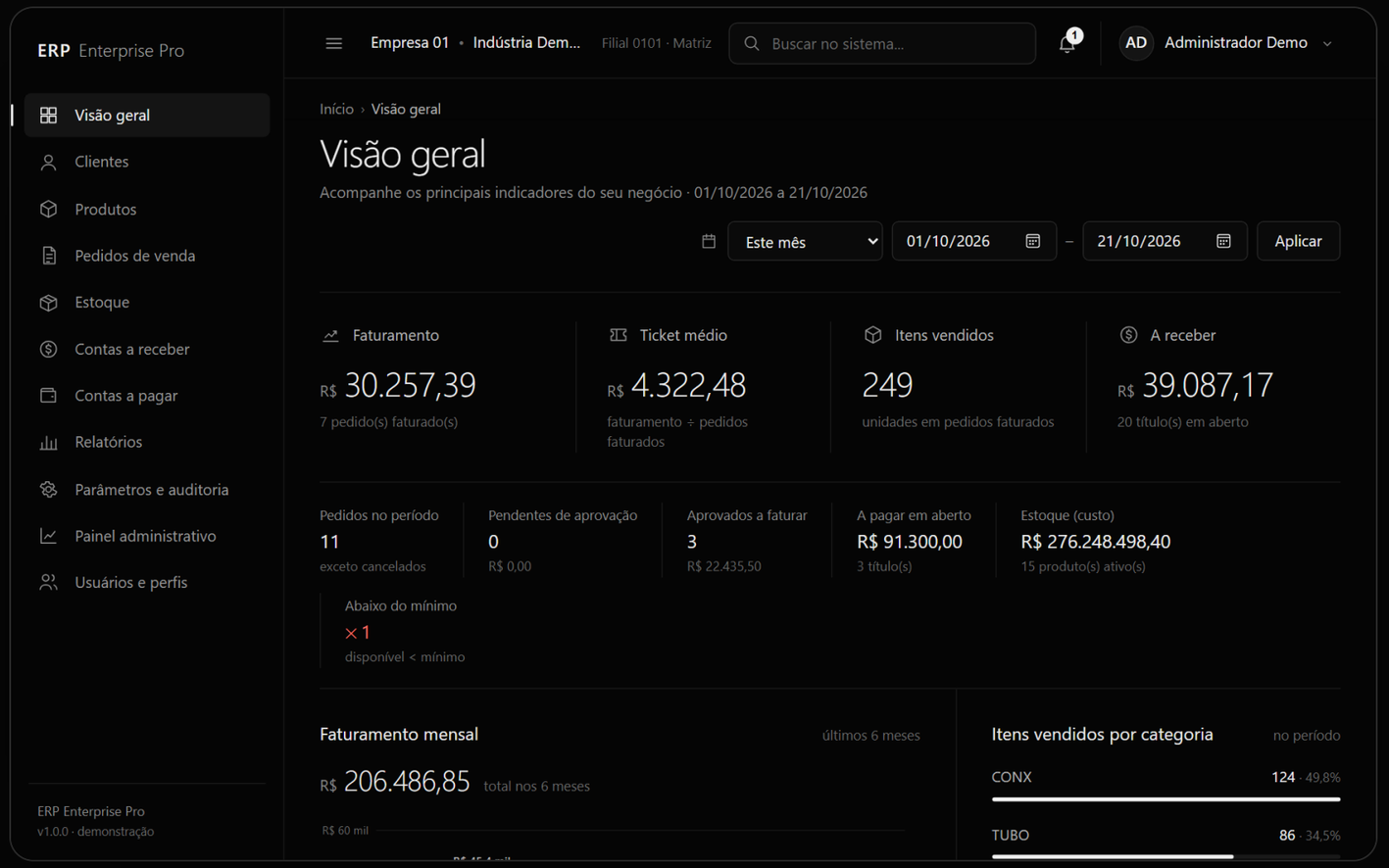
The first screen after sign-in: key business figures, calculated live.

1. **Period**
   This month, last month, last 30 days, year or custom dates.

2. **Indicators**
   Revenue, average ticket, items sold and accounts receivable.

3. **Charts**
   Revenue for the last 6 months and items sold by category.

4. **Attention**
   Pending items, stock below minimum and latest orders.



Master data

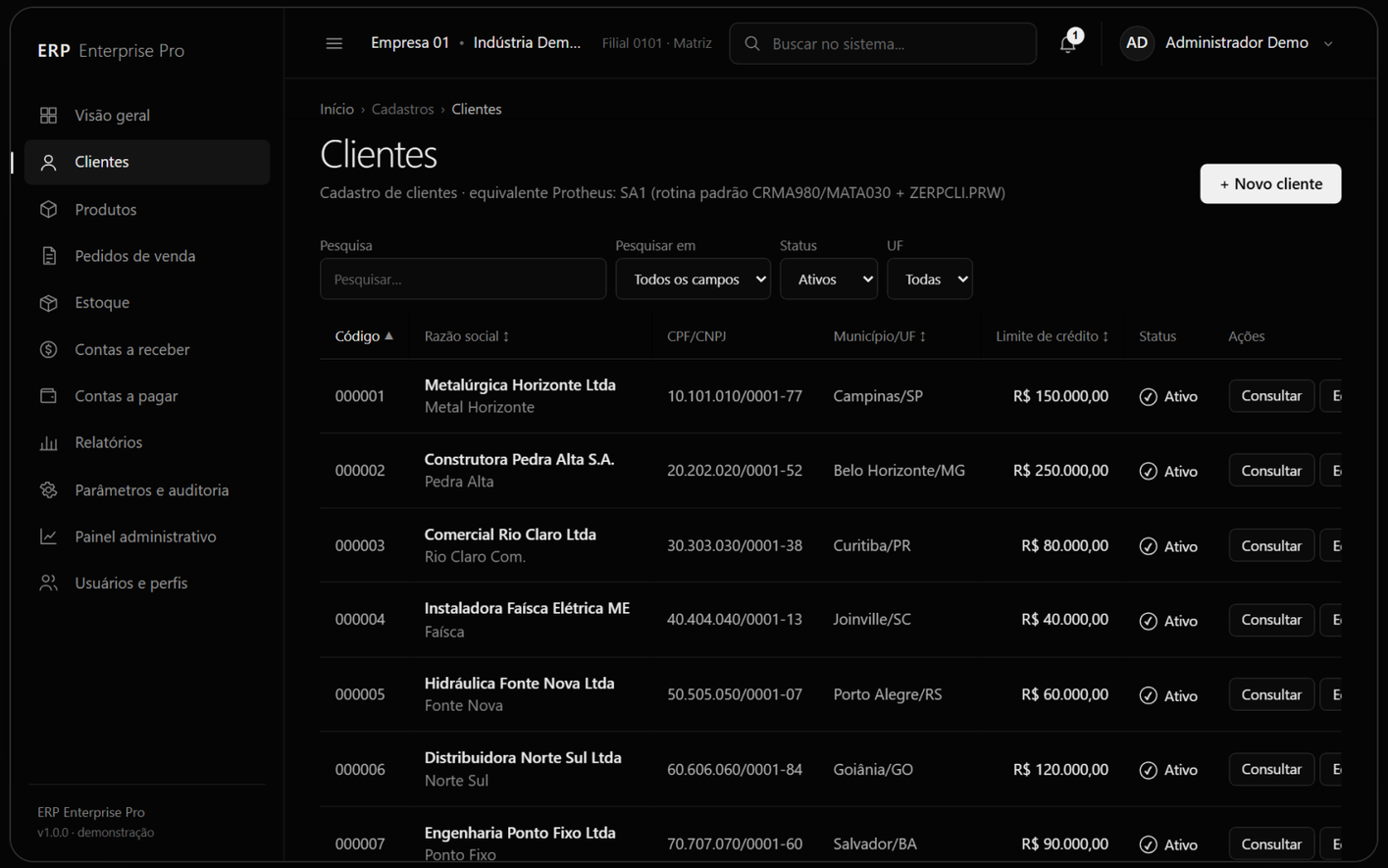
Customers and products, with search, filters and change history.

1. **Customers**
   Brazilian tax ID (CPF/CNPJ) validated for check digits and duplicates, automatic code, credit limit.

2. **Products**
   Price, cost, group, minimum stock and an alert when stock runs low.

3. **History**
   Each change stores the old and new value, with user and date.

4. **Global search**
   The search field at the top jumps straight to customers, products or orders.



Sales order

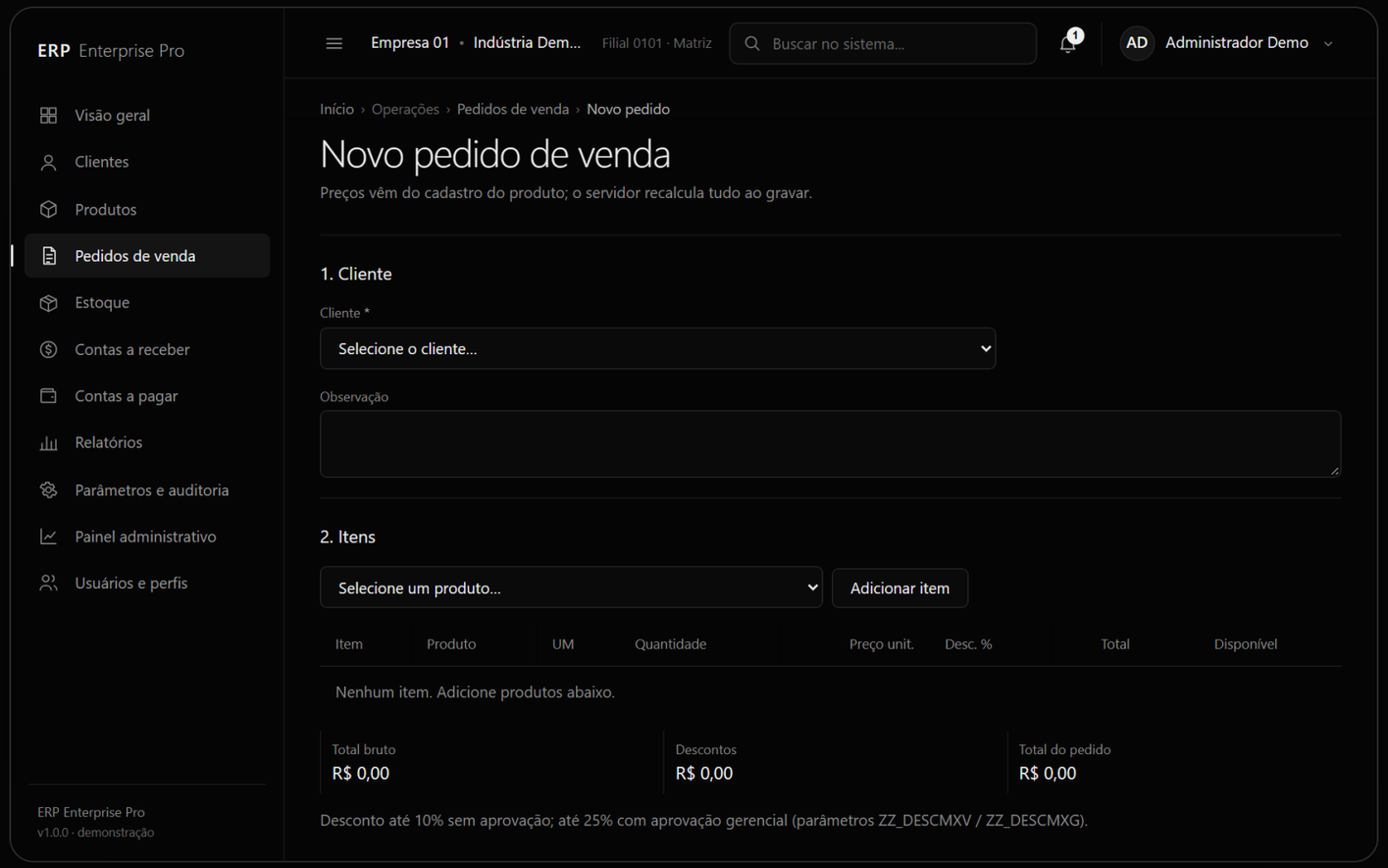
The sales rep builds the order; the system calculates and checks everything before submitting.

1. **Customer**
   Available credit appears as soon as the customer is selected.

2. **Items**
   Product, quantity and per-item discount, with the stock available.

3. **Totals**
   Gross, discounts and net calculated on screen and recalculated on the server.

4. **Save**
   As a draft, or submitted straight for approval.



Approval and invoicing

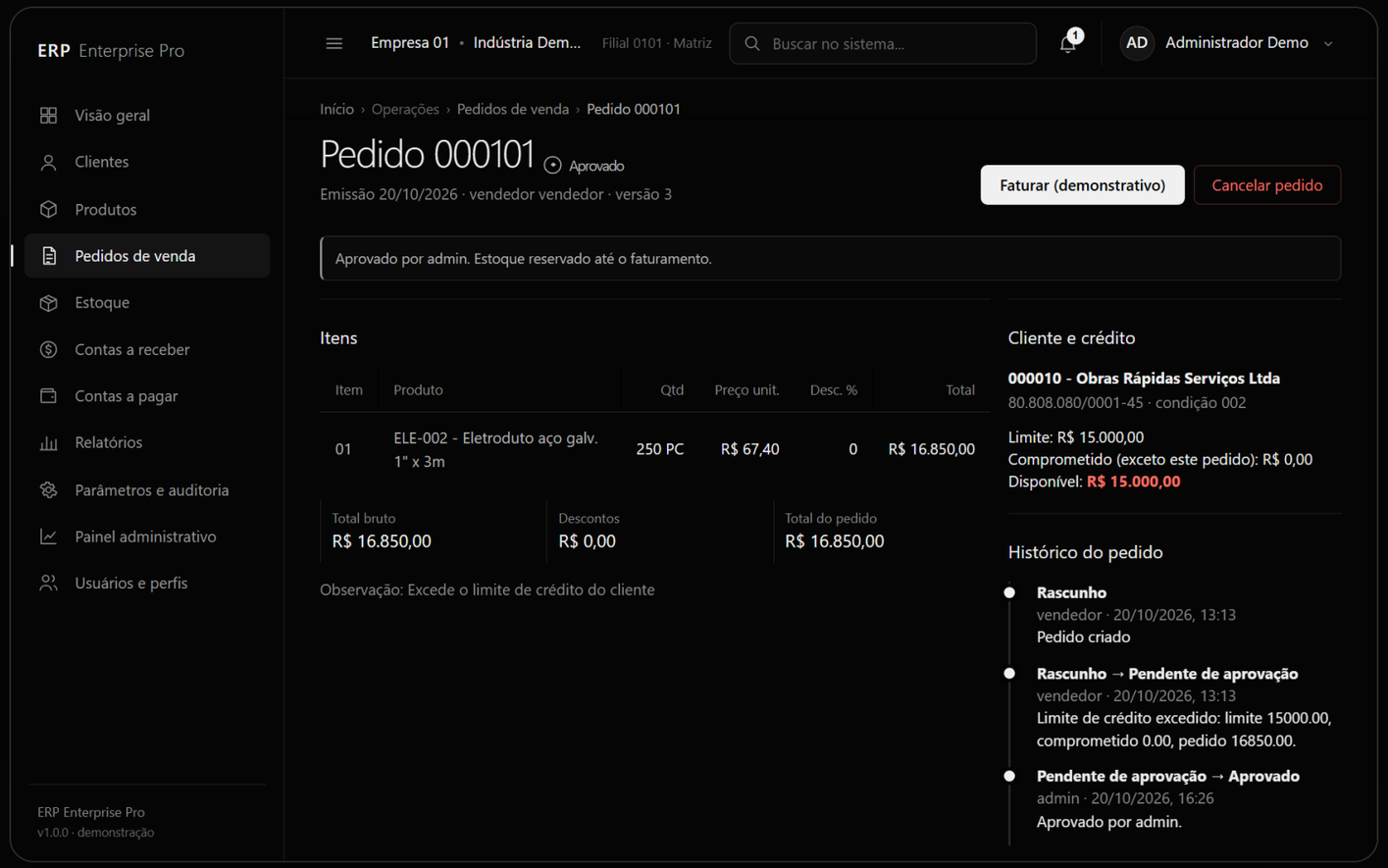
Each status enables only the buttons allowed at that point.

1. **Approve or reject**
   The manager decides on out-of-policy orders, with the reason logged.

2. **Invoice**
   Deducts the reserved stock and creates the receivables.

3. **History**
   A timeline with every status change.

4. **Note**
   Demo invoicing: does not issue NF-e (Brazilian e-invoice) or calculate taxes.



Inventory

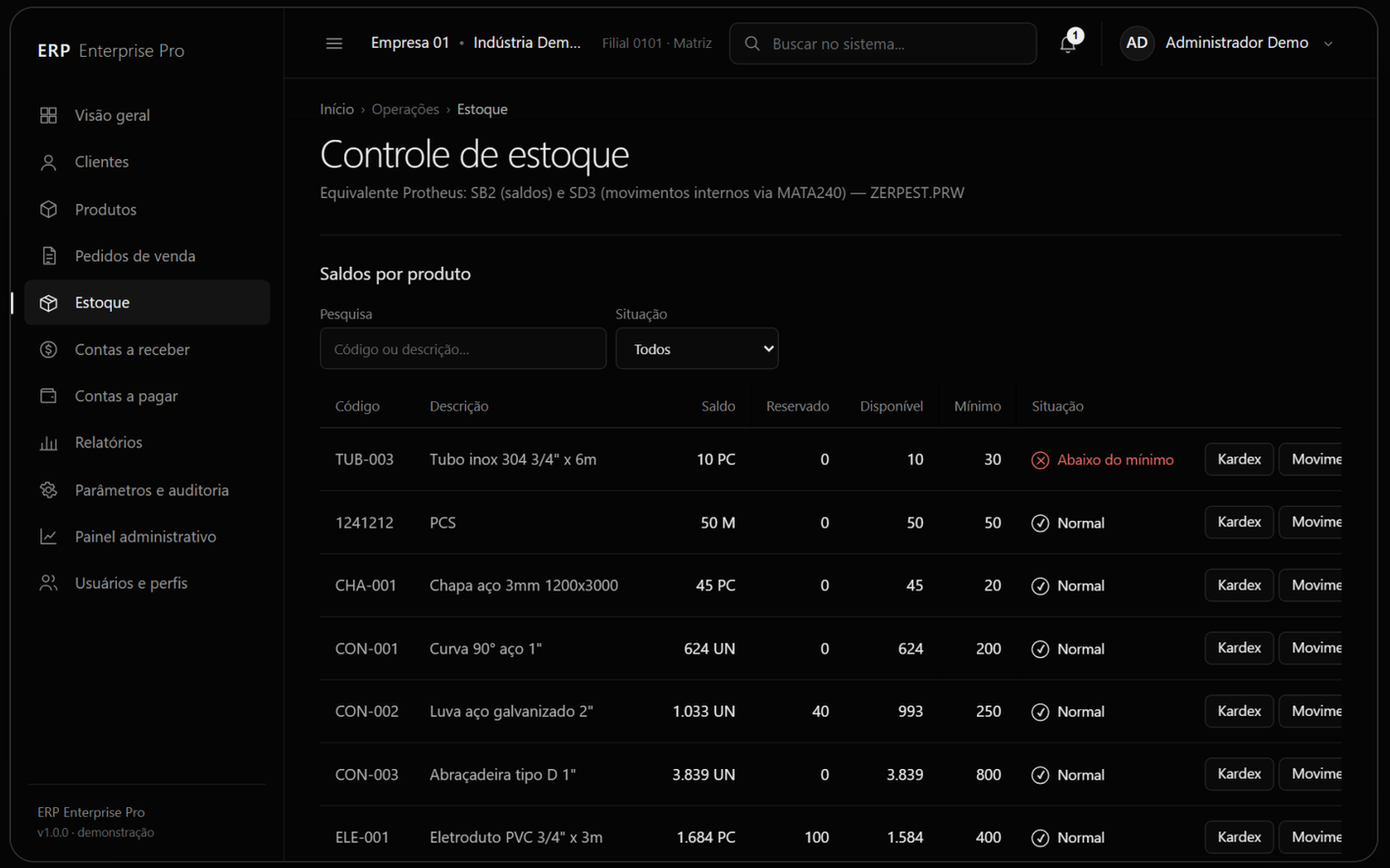
On-hand, reserved and available quantity for each product, with movement history.

1. **Available**
   On-hand minus what is already reserved for approved orders.

2. **Stock movements**
   Manual receipts and issues with a reason (purchase, adjustment, loss).

3. **Stock ledger (Kardex)**
   Every receipt and issue for a product, in order.

4. **Alert**
   Products below minimum stock are highlighted.



Finance

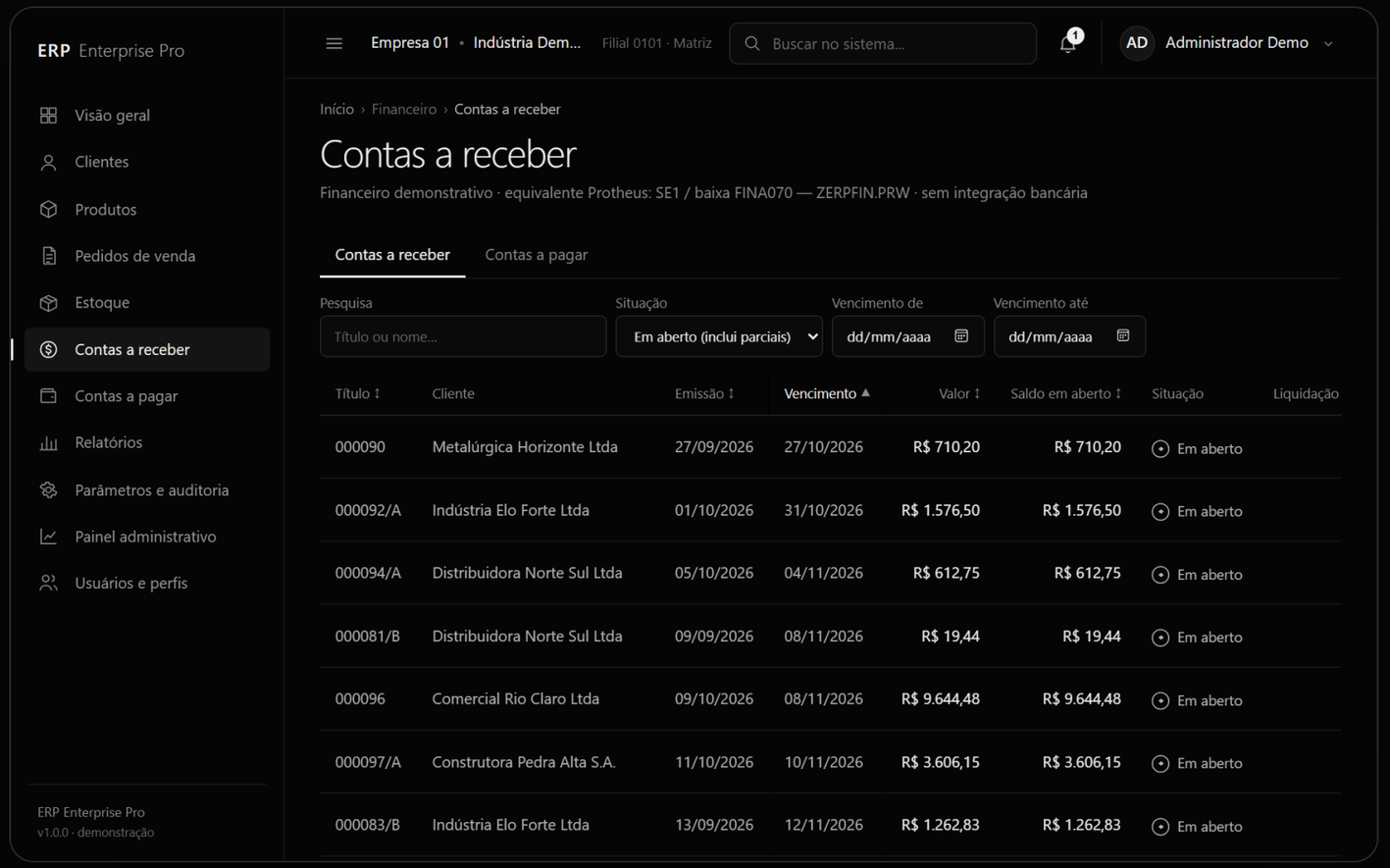
Accounts receivable (created at invoicing) and accounts payable.

1. **Status**
   Open, partial, overdue or settled — shown with a symbol and text.

2. **Settle**
   Records full or partial payment, with date and amount.

3. **View**
   Shows the item and every payment already recorded.

4. **Alerts**
   Overdue items show up in notifications.



Reports

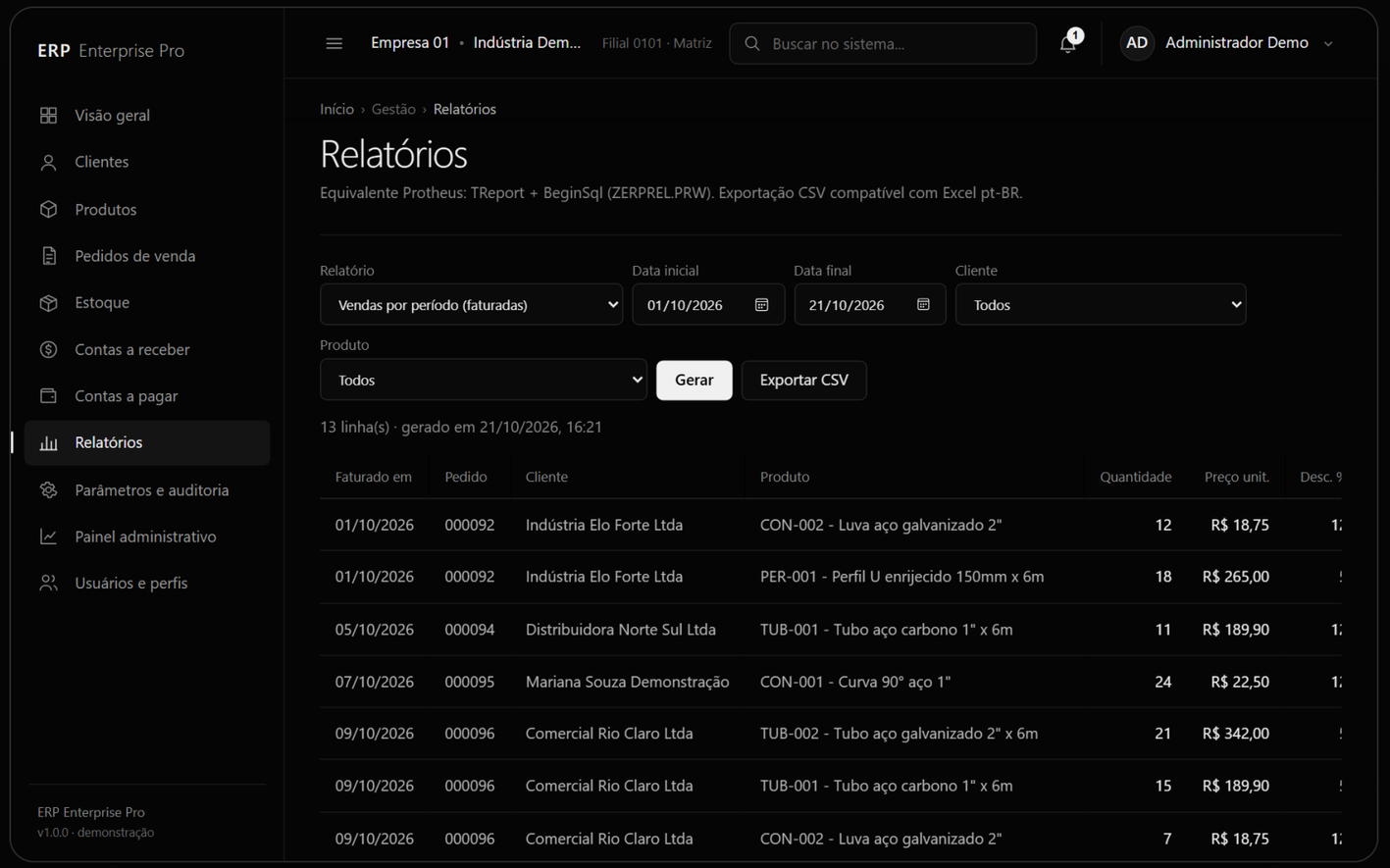
Six reports with filters and export to Excel.

1. **Choose**
   Sales, inventory, finance and more.

2. **Filter**
   Period, customer and product, depending on the report.

3. **Generate**
   Results on screen, with totals for value columns.

4. **Export**
   CSV file that opens directly in Excel (Brazilian locale).



Administration

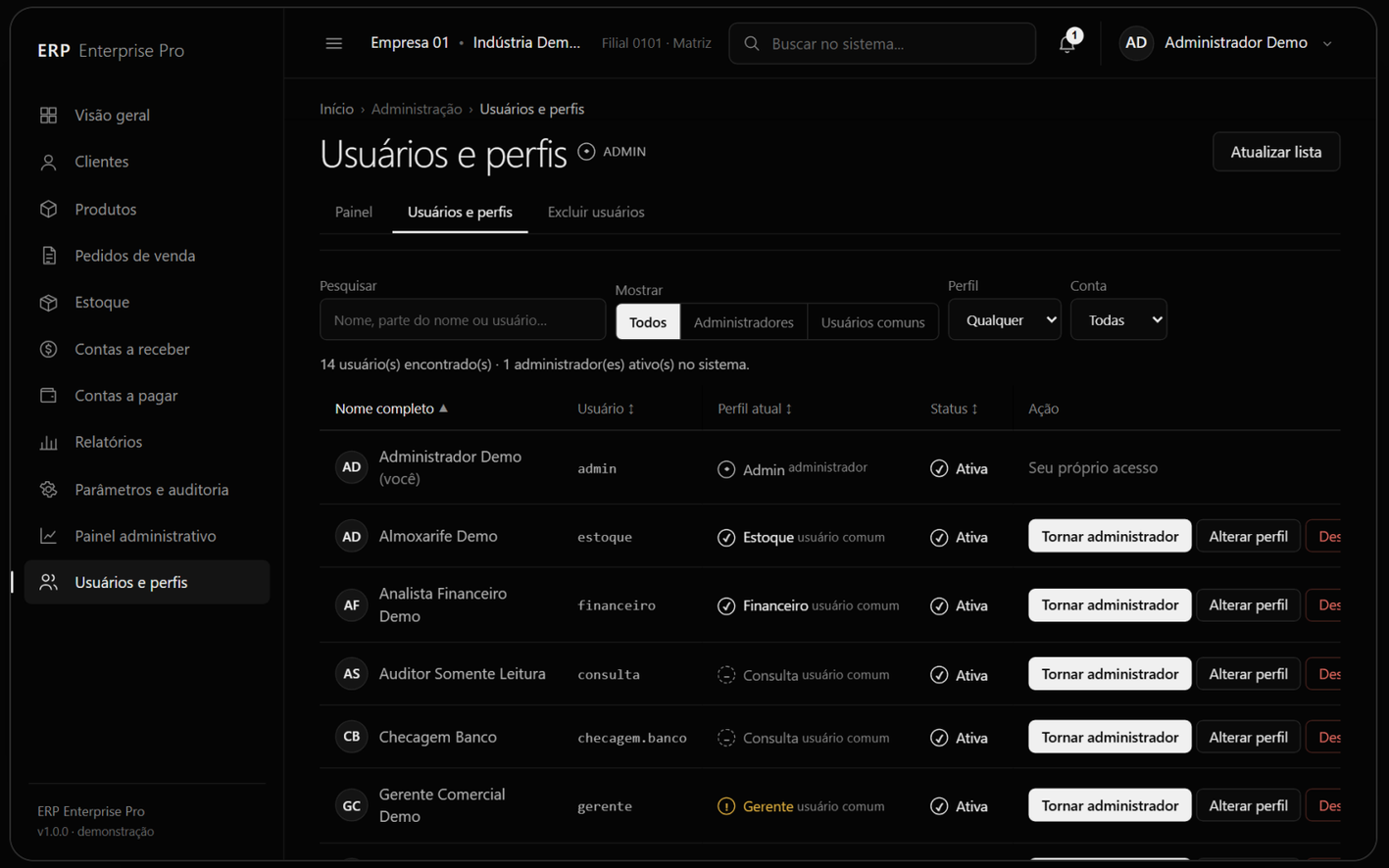
Administrator-only area: users, profiles and a management view.

1. **Users and profiles**
   Change a user's profile; grant or revoke administrator access.

2. **Accounts**
   Deactivate, reactivate and delete deactivated accounts.

3. **Dashboard**
   Sales, margin, active users and denied access attempts.

4. **Traceability**
   Every profile change is logged with who made it and when.



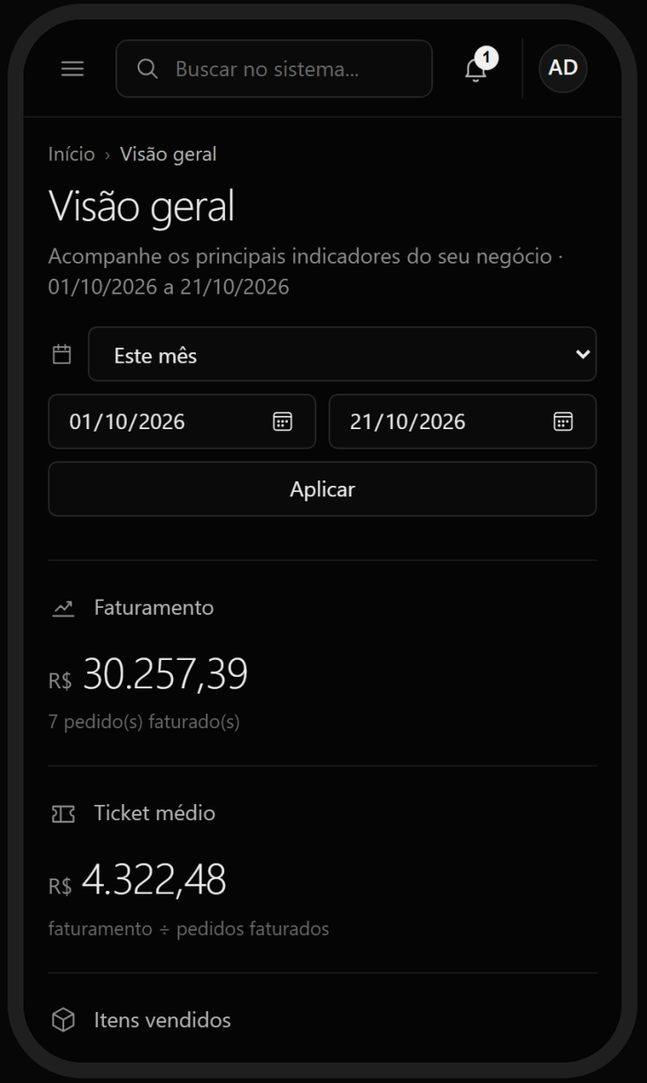
Works on mobile too

The same screen reflows into a single column. No page needs horizontal scrolling.

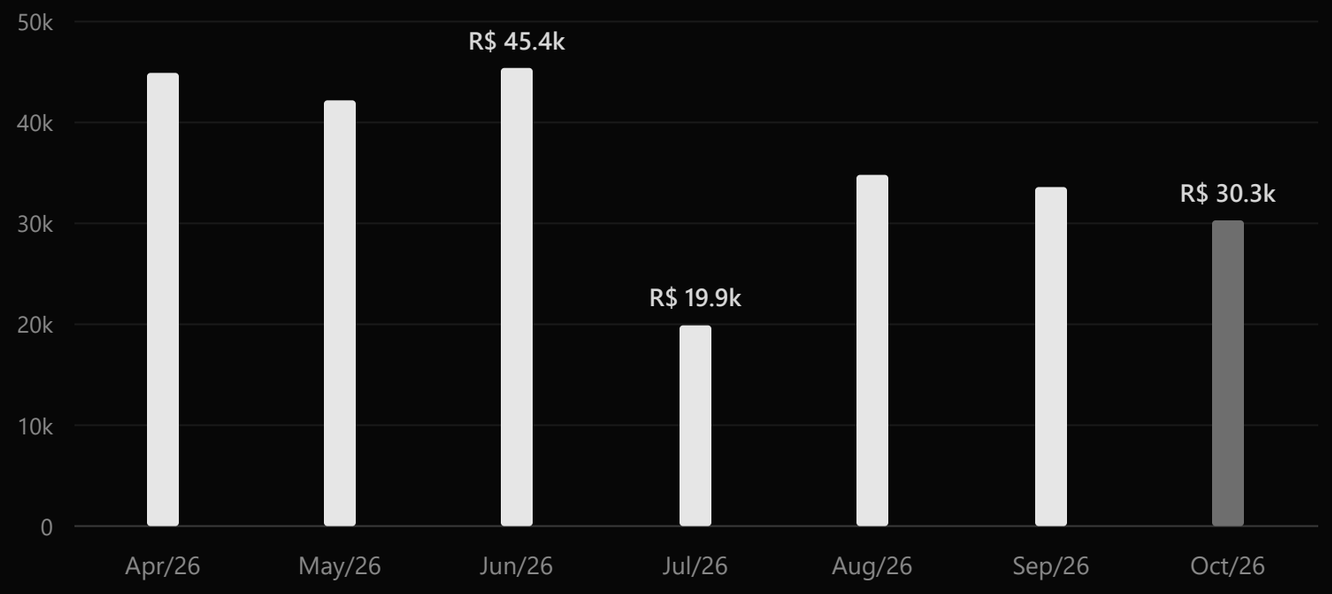
1. **Same address**
   Just open the system in the phone's browser, on the company network.

2. **Collapsed menu**
   The menu opens from the button at the top and stays out of the way.

3. **Keyboard**
   Every command also works from the keyboard (Tab, Enter, Esc).



Monthly revenue



Invoiced orders per month. Grey bar: October is partial, through Oct 21. Fictitious demo data.

Total invoiced (Apr–Oct)

R$ 251,498.74

Invoiced orders

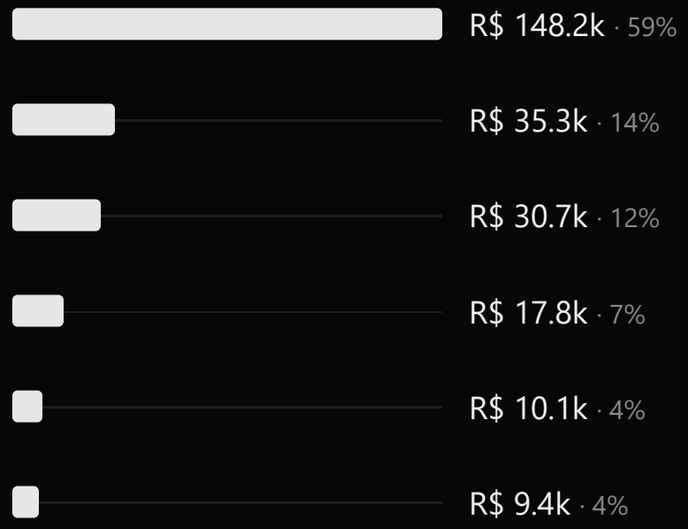
81

Average ticket

R$ 3,104.92

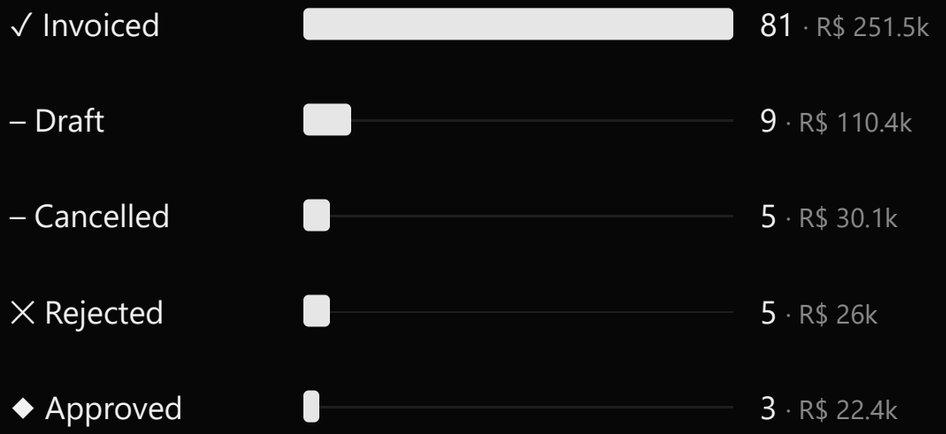
Where revenue comes from and how orders are doing

Revenue by product group



Pipes account for more than half of invoiced revenue.

Orders by status



81 of 103 orders (79%) have already been invoiced.

Security from the start

Protected passwords

Stored with strong hashing (scrypt). Not even the administrator can read them.

Secure sessions

Sessions expire after 8 hours; the cookie is out of reach of scripts; signing out ends the session on the server.

Brute-force protection

5 wrong passwords lock sign-in for 60 seconds.

Server-side permissions

Every action is checked against the user's profile, even if someone bypasses the screen.

Tamper-proof audit log

Who did what, and when. Records cannot be edited or deleted.

No partial writes

If something fails in the middle of an operation, everything is rolled back automatically.

Automated tests

Tests by area



Result

49/49

passed

The tests simulate real use through the API: creating, approving and invoicing orders, attempting forbidden access, sending invalid data and even forcing a failure in the middle of a write.

They run in under 20 seconds with `testar.cmd`.

Technical self-assessment

Score by criterion (points earned / points possible)



Self-assessment recorded in docs/CONFORMIDADE.md (Oct 20, 2026). It should be reviewed by a Protheus developer.

Overall score

77 / 100

solid foundation, with points to review

The biggest deduction is in ADVPL: the source files have not been compiled, so only a review in a staging environment can confirm they are correct.

Limitations and next steps

What this version does not do

- Does not issue NF-e or calculate taxes (demo invoicing).
- Not integrated with Protheus; uses its own database.
- No banking integration (boleto, CNAB).
- No HTTPS: built for local or internal-network use.
- ADVPL source files not compiled yet.

Suggested next steps

1. Compile and test the ADVPL source files in staging (never in production).
2. Validate tax rules (TES, customer type) with the tax team.
3. Define the official credit policy with finance.
4. Deploy with HTTPS for access outside the network.
5. Gather feedback from users in each area.

ERP Enterprise Pro

Thank you. Questions?

HOW TO OPEN

Run `iniciar.cmd` and go to `http://localhost:3000`

DEMO USERS

admin, gerente, vendedor, estoque, financeiro, consulta · password `Demo@2026`

ILLUSTRATED GUIDE

18 step-by-step images in `docs/guia`